Ashritha Gonuguntla

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Education

Carnegie Mellon University, School of Computer Science Pittsburgh, PA
Master of Software Engineering, **GPA 4.0/4.0** Aug 2025 – Dec 2026
- Coursework: AI Agents, Deep Learning Systems, Generative AI, Multimodal ML, Diffusion and Flow Matching, ML in Production, Computer Systems, Software Architecture, Mobile Robots, F1-Tenth Autonomous Racing.
- Teaching Assistant, four courses: ML in Production (Prof. Kästner), Prompt Engineering (Prof. Breaux), Computer Systems (Prof. Kesden), Engineering Computation (Prof. Yamakawa).

K L University Andhra Pradesh, India
B.Tech Computer Science, AI & Robotic Process Automation, **GPA 9.84/10, rank 2** Jun 2018 – Mar 2022
- Academic Excellence Award for second rank in the graduating class.

Experience

ePart Services (CMU MSE Studio client), Technical Lead, ML & Architecture 2026 – Present
- Leading **5 engineers** shipping a confidence-routed attribute-prediction service to the client: BGE+FAISS retrieval, a tiered rule engine, and Mahalanobis confidence routing over 4.08M attribute rows; 95.74% ProductType accuracy on 12,958 held-out products.
- Caught a deployment-blocking evaluation error: semantic-only scoring capped fused confidence at 0.30, making the **0.85 auto-accept gate mathematically unreachable**; added end-to-end rule composition and per-ProductType calibration before rollout.
- Hardened the Python 3.12/FastAPI service with ruff, black, mypy, **289 tests, and an 85% branch-coverage gate**; load-tested Prometheus-instrumented serving at 50 req/s against a 200 ms p95 SLO and isolated p95 latency to the encoder (∼38 ms).

Cisco, Software Engineer → Senior Software Engineer Aug 2022 – Jul 2025
- Tech-led 9 engineers across Cisco Secure Firewall's two telemetry paths in C++/Go: a gRPC plane with control/data channels, protobuf metrics, mTLS, ACK-based cancellation, plus gNMI/OpenConfig YANG subscriptions, scale-tested to 1,000 collector endpoints; the feature carries **$100M in ARR**.
- Rebuilt the intrusion-event stream-processing and query path with tailored indexing and range-based partitioning: **40% lower latency at 2x telemetry volume**; mentored 4 engineers through design and performance reviews.
- Cut median SOC triage time **57%** with an agentic RAG incident assistant: multi-step LLM orchestration and tool calling over firewall telemetry and configuration, OpenSearch vector retrieval on AWS SageMaker/Lambda, root-cause summaries.
- Contributed to **LLM post-training** for a 1B security model, later released as Antares-1B: prepared CVE/CWE SFT data, evaluated early checkpoints, built network-disabled Docker sandboxes for read-only repo exploration; scope preceded GRPO.
- Eliminated duplicate job executions across distributed schedulers by architecting a **Kafka-based asynchronous processing framework** for batching, deduplication, and coordination of job requests; removed the scheduler as a scaling bottleneck.
- Carried **production on-call** for telemetry and observability services: alert triage and log/query diagnosis through mitigation and follow-up fixes across ingestion and operator-facing detection workflows.

Cisco, Software Intern Jan 2022 – Jun 2022
- Built cloud-native security observability on Kubernetes with **Prometheus, Grafana, CloudWatch**; automated AWS provisioning with Terraform, Ansible, CloudFormation.

Adobe Research, Research Intern May 2021 – Aug 2021
- Co-authored Outage-Watch (ACM ESEC/FSE'23), a mixture density network for QoS tail-risk forecasting: **AUC 0.98, MTTD cut up to 88%**; co-inventor on US Patent App. 17/656,263.

IIIT Hyderabad, Research Intern, Speech and Vision Lab Apr 2021 – May 2021
- Implemented neural architectures for automatic speech recognition and text-to-speech synthesis in **TensorFlow and PyTorch**.

University of Kerala, ML Intern, Computational Biology & Bioinformatics Oct 2020 – Dec 2020
- Built a **CycleGAN** pipeline for cross-domain style transfer, enhancing structural patterns in cellular microscopy images.

Microsoft, Software Engineering Intern May 2020 – Aug 2020
- Built the metrics and insights layer of **Viva Insights** over terabytes of data for 27K+ companies, in Azure and **PySpark**; validated the influence and network algorithms end to end for customer release.

Publications

- *The Replay Gap: Static Evaluation of Model Switching in LLM Agents Scores the Wrong World* (sole author). **COLM'26 Workshop on Efficient Reasoning.** Branching rollouts over ∼900 SWE-bench trajectories show static replay misscores 92–97% of switching decisions; serving-stack cost and determinism characterized over 20,141 requests.
- *The Plan, Not the Decoder: Diagnosing and Repairing Compositional Failure in Reasoning-Augmented Text-to-Image Generation* (sole author). **ECCV'26 MUCG Workshop, Oral.** Localized compositional failure to the planner rather than the decoder; training-free repair gained +10.7 points over ∼12K images.
- *APIShift: Verifiable Agent Learning for API Contract Migration.* **EMNLP 2026, under review.** A verifiable RL environment over 1,549 real API version pairs from 449 providers; GRPO and LoRA training on Qwen2.5-3B with a documented reward-variance collapse and its repair directions.
- *Resolving Spatial and Semantic Conflicts: Multi-LoRA Composition via Softmax Routing and Cross-Adapter Attention Conditioning.* **AAAI'27, under review.** Training-free softmax noise routing and cross-adapter attention: +6.8% CLIP-T, +16.7% CLIP-I.

Projects

nanoinfer | *C++20, ARM NEON, CUDA, cuBLAS/cuDNN, ONNX Runtime* Code
- Built a C++20 edge inference engine (im2col+SGEMM, NEON depthwise, fused epilogues, liveness-planned arena with no forward-pass allocations): **0.99x ONNX Runtime** on the dense CNN, 0.60–0.77x on depthwise, 37–77% less activation memory.

replay-gap | *vLLM, FP8/AWQ, SWE-bench, mini-SWE-agent* COLM'26 Workshop, sole author
- Built a resumable branching-rollout harness that forks live SWE-bench agent trajectories, rebuilds each Docker environment, and continues the fork under a different model: **889 rollouts** over 20,141 requests and 42.5 GPU-hours.

APIShift | *GRPO, LoRA, Qwen2.5-3B, verifiable rewards, RTX 3090* EMNLP 2026, under review
- Built a verifiable RL environment for API contract migration over **1,549 real API version pairs** from 449 providers and ten breaking-change types: a manager emits JSON actions to Diff, Patch, Test and Rollback specialists under a five-component reward.

batch-invariance | *vLLM 0.28, Qwen3-1.7B, RTX 4090, CUDA graphs* Code
- Built a probe harness isolating batch size, batch composition, repeat rate, and throughput against vLLM 0.28, and priced the determinism fix at **54 to 67%** of throughput, a cost the documentation states but never quantifies.

specdec | *PyTorch, Qwen3-0.6B draft, Qwen3-4B target, RTX 4090* Code
- Implemented speculative decoding from scratch and showed it **loses at every window size** in eager PyTorch (0.87x to 0.51x): the draft costs 0.649 of a target pass, so break-even reduces to acceptance = cost ratio.

Query-Rewrite Evaluation | *Qwen3-1.7B, LoRA (PEFT), BM25, FastAPI, QReCC, nDCG, Kendall τ* Code
- Fine-tuned Qwen3-1.7B with LoRA (PEFT, prompt-masked loss, cosine schedule) on 48,415 QReCC pairs, lifting nDCG@10 from 0.400 to **0.481** against a 0.505 human reference; saved 7 checkpoints as evaluation systems.

Gated Residual RL | *MuJoCo, MetaWorld, PPO, SAC, PETS/MPPI* Code
- Caught a policy scoring **85% in metrics but 0% on camera**, traced it to seed and protocol leakage, and corrected the evaluation before reporting.

mlops-replay | *Python, paired bootstrap, drift monitoring, Docker* Code
- Replayed **120 months** of production-style data through score/label/retrain/gate/promote; a paired-bootstrap gate shipped 3 regressions vs. 22 for monthly retraining.

GridPilot | *Python, pandapower, IEEE 118-bus, LLM tool use, FastAPI* Code
- Built a tool-using LLM grid operator and 50-incident benchmark; a code-enforced **simulate-before-commit guardrail** cut load lost **32%** vs. no action with 3 invalid calls across 264 turns.

F1TENTH Autonomous Racing | *ROS2 Humble, Python, Jetson, Hokuyo LiDAR* CMU 16-663, team
- Built reactive navigation on a physical 1/10-scale racecar: iTTC emergency braking, PID wall-follow, disparity-extended Follow-the-Gap, adaptive Pure Pursuit, RRT*. **Won 3 of 3 races.**

SiMATU: LLM Unlearning | *Phi-3.5-mini 3.8B, RWKU, causal tracing* CMU Deep Learning Systems, team
- Made a 3.8B model forget targeted facts: knowledge sits in early layers but only late-layer edits remove it, so a three-layer edit hit **91.9% unlearning** at 97.4% retention, up from 74.1%.

Autonomous Robotic Arm for EV Charging | *computer vision, 6-DOF manipulation, team lead* K L University
- Led a team building the perception pipeline for a **6-DOF robotic arm** performing autonomous EV charger docking: object detection and spatial reasoning to localise the charging port.

Federated Fault Diagnosis | *PyTorch, Flower, Ray, CWRU bearing data* Code
- Trained a 1D CNN on **10 federated clients** with mismatched fault distributions and late reporters: rounds to 90% accuracy fell 47 to 21 at 96.5%±3.5; FedProx, sold for exactly this, did not beat the baseline.

Honors and Awards

- **Mary Shaw Award for Creativity in Software Engineering**, Carnegie Mellon, 2026 · 5th place, Gray Swan AI red-teaming challenge, 2026 · CMU Moon Miners, NASA Lunabotics, 2026.
- **JN Tata Scholar** and **KC Mahindra Scholar**, 2025 · 2nd place, Best Adoption category, AWS Graviton Challenge, 2024 · Recognised under Cisco's Achieve Operational Excellence category.
- **Winner, Adobe SheCodes**, 2020 · **National winner, Smart India Hackathon**, 2020 · McKinsey Next Generation Women Leaders, 2021 · Walmart CodeHers finalist, 9th rank, 2021 · Cisco Ideathon finalist, 2021.
- **Gold Medalist and Best Outgoing Student**, CBSE secondary school board, class of 2015.

Leadership and Service

- **Regional Chapter Lead, GirlScript Foundation India**: led the Andhra Pradesh chapter of a nonprofit running free coding programs for women and underrepresented students.
- **Volunteer, eVidyaloka Trust and Empower Ananya**: digital literacy for students in underserved rural and urban communities across India, through video classrooms and outreach.
- **Honored Speaker, ShaktiCon 2023**: "5G IoT Security: Protecting the Next Generation of Connected Devices." · MSE Leadership Initiative, Carnegie Mellon.
- **Co-founder, Vector AI** (2019–2020): a cultural gaming startup; shipped *Ethan's Adventure*, a puzzle game with custom enemy and character AI, to the Google Play Store.

Technical Skills

Languages: Python, C++20, Go, Java, C, SQL, Bash, TypeScript/JavaScript.
Additional tooling: TensorFlow, scikit-learn, Ray, Flower, MLflow, PySpark, Redis, MongoDB, Oracle, Splunk, Helm, Terraform, Ansible, GCP, Azure, Isaac Sim, Gazebo, Nav2, GoogleTest.